INCH-POUND

MIL-R-44277A(GL)
15 November 1990
SUPERSEDING
MIL-R-44277(GL)
30 September 1987

MILITARY SPECIFICATION

RATION, COLD WEATHER, ASSEMBLY OF

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the assembly requirements for the Ration, Cold Weather.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

A-A-20135	-	Sugar, Refined and Sugar, Brown
A-A-20175	-	Chewing Gum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8970

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A-A-20177	-	Candy and Chocolate Confections
L-F-560	-	Fork, Knife and Spoon, Picnic Plastic
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
EE-M-101	-	Matches, Safety: and Matches, Nonsafety (Strike-Anywhere)
QQ-A-1876	-	Aluminum Foil
HHH-C-575	-	Coffee, Instant
MMM-A-250	-	Adhesive, Water-Resistant (for Closure of Fiberboard Boxes)
PPP-T-76	-	Tape, Pressure Sensitive Adhesive, Paper (for Carton Sealing)
PPP-T-97	-	Tape, Packaging/Industrial Filament Reinforced
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-C-3031	-	Cocoa Beverage Powder
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-43338	-	Cream Substitute, Dry, Non-Dairy
MIL-P-43988	-	Paper, Toilet, Tissue (Packet in Sheet Form)
MIL-C-44072	-	Cookies, Oatmeal; and Brownies; Chocolate Covered
MIL-B-44135	-	Beverage Base: Orange Compressed - (Operational Ration Component)
MIL-O-44136	-	Oatmeal Cookies and Granola Bars - (Operational Ration Component)
MIL-C-44137	-	Chicken A La King, Cooked, Dehydrated - (Operational Ration Component)
MIL-C-44138	-	Chicken and Rice, Cooked, Dehydrated - (Operational Ration Component)
MIL-V-44139	-	Vermcelli with Meat and Sauce, Cooked, Dehydrated (Operational Ration Component)
MIL-B-44140	-	Beef Stew, Cooked, Dehydrated - (Operational Ration Component)
MIL-E-44141	-	Escalloped Potatoes with Pork, Cooked, Dehydrated (Operational Ration Component)
MIL-C-44148	-	Chicken Stew, Cooked, Dehydrated - (Operational Ration Component)
MIL-P-44276	-	Packaging of Ration, Cold Weather Components (in Pouches)
MIL-O-44278	-	Oatmeal, Instant (Operational Ration Component)
MIL-F-44279	-	Fruit Bars - (Operational Ration Component)
MIL-N-44280	-	Nut Raisin Mix - (Operational Ration Component)
MIL-B-44281	-	Beverage Base, Powder, Apple Cider, Natural and Artificial Flavored (Operational Ration Component)
MIL-S-44283	-	Soup, Instant, Reduced Sodium, Chicken Noodle
MIL-T-44284	-	Tea, Instant, Sweetened, Lemon Flavored (Operational Ration Components)

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STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1238 - Flow Rates of Thermoplastics by Extrusion Plastometer
D 1505 - Density of Plastics by Density Gradient Technique
F 88 - Seal Strength of Flexible Barrier Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

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(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless specific exemption has been obtained.

3. REQUIREMENTS

3.1 Components.

3.1.1 Ration bag. Ration bag shall be made from polyethylene tubing that is approved for food use and is 0.008 to 0.010 inches thick from a blend of not less than 50 percent linear low density polyethylene and polyethylene. The linear low density polyethylene portion of the blend shall be the copolymer of ethylene and octene-1 having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922g/cm^3 in accordance with ASTM D 1505. The color range shall be from 37778 through 37886 of FED-STD-595. The inside dimensions of the bag shall be 7-1/8 inches wide and 11-7/8 inches long (± 1/8 inches). The bag opening shall be across the width. The bag manufacturer's seal shall be a minimum 1/8 inch wide and located with the outside edge of the seal 1/2 inch (± 1/8 inch) from one end of the bag. The seal shall form a hermetic closure. The average seal strength of each seal shall not be less than 9 pounds per inch of width and no individual specimen shall be less than 7 pounds (see 4.2.1.2). A 1/4 to 3/8 inch deep slit or V-shaped tear notch shall be located at one end of the bag within 1/2 inch of the longitudinal center line of the bag. The bag shall be marked A or B (see table III and 5.4.1).

3.1.2 Accessory packet. The accessory packet shall be fabricated in accordance with 3.1.2.1 or 3.1.2.2.

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3.1.2.1 Preformed packet. The preformed accessory packet shall be fabricated from material made by hot-extruding or laminating a minimum of 0.0015 inch thick (- 10 percent) polyethylene on 0.00035 inch thick (± 10 percent) aluminum foil and laminating the opposite side of the foil to 0.0005 inch thick (± 10 percent) polyester. The color of the exterior (polyester) side of the laminate shall conform to numbers 37778 through 37886 of FED-STD-595. Any difference in gloss caused by the polyester exterior layer will be considered acceptable. The inside dimension of each packet shall be 4-1/2 inches wide by 7 inches long (± 1/8 inch). Each packet shall be heat sealed on three edges with 3/8 inch wide (± 1/8 inch) seals. A 1/8 inch wide (± 1/16 inch) lip may be incorporated at the open end of the packet to facilitate opening and filling of the packet. A tear nick or notch shall be located on the outer edge of one or both side seals to facilitate easy opening of the packet. Each tear nick or notch shall be a minimum 1/16 inch deep, but shall not extend into the side seal to a depth that reduces the seal width to less than 1/8 inch. The average seal strength of the packet seals shall not be less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width (see 4.2.1.2).

3.1.2.2 Form-fill-seal packet. The form-fill-seal accessory packet shall consist of a tray-shaped body with a tray-shaped or flat sheet heat sealable cover. The tray-shaped body and tray-shaped cover (when used) shall be formed by drawing flexible laminated material into an appropriately shaped cavity. The tray-shaped or flat sheet cover shall be applied by a continuous hermetic heat seal along the entire perimeter of the tray-shaped body. The tray-shaped body and tray-shaped cover (when used) of each packet shall be fabricated from 0.0015 inch thick (± 10 percent) linear low density polyethylene bonded to 0.0005 inch thick (± 10 percent) aluminum foil with 7 pounds per ream low density polyethylene extrusion resin and 0.00075 inch thick (± 10 percent) oriented polypropylene bonded to the opposite side of the aluminum foil with 10 pounds per ream low density polyethylene extrusion resin. The flat sheet cover (when used) of each packet shall be fabricated from 0.0015 inch thick (± 10 percent) linear low density polyethylene bonded to 0.00035 inch thick (± 10 percent) aluminum foil with 7 pounds per ream low density polyethylene extrusion resin and 0.0005 inch thick (± 10 percent) polyester bonded to the opposite side of the aluminum foil with 10 pounds per ream low density polyethylene extrusion resin. The linear low density polyethylene shall be the copolymer of ethylene and octene-1 having a melt index range of 0.8 to 1.2 g/10 minutes in accordance with ASTM D 1238 and a density range of 0.918 to 0.922 g/cm^3 in accordance with ASTM D 1505. The color of the exterior side of the laminates (polypropylene side of the tray-shaped body and tray-shaped cover and the polyester side of the flat sheet cover) shall be as specified in 3.1.2.1. The outside dimensions of each packet shall be 5-1/4 inches wide by 7-3/8 inches long (± 1/16 inch). The closure seal at the perimeter of the packet body and cover shall be a minimum 1/4 inch wide. A tear nick or notch shall be located on one or both outside edges of the packet length or width to effect easy opening of the packet in the machine direction of the packet body and cover laminates. Each tear nick or notch shall be a minimum 1/16 inch deep, but shall not extend into the seal to a depth that reduces the seal width to less than 1/8 inch. Alternatively, if the packet has serrated edges, the serrations may be used as tear notches provided

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that the serrations are located to effect easy opening in the machine direction of the packet laminates, that the serrations are sharp (no plastic tailings exist), and that the serration depth and minimum seal width at the serrations are as specified for the tear nick or notch. The minimum seal strength of the packet seals shall not be less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width when tested in accordance with 4.2.2.2. Alternatively, the form-fill-seal accessory packet may be made on equipment that does not change the shape of the bag laminate material from the flat sheet form. The body shall not be formed into a tray, but shall be in the form of a flat sheet-like the cover. When this equipment is employed, the packet shall be made in accordance with the requirements of 3.1.2.1, except that all four sides of the packet bag shall be sealed in the sequence required by the equipment used. Projection of the packet material beyond the outside edges of the seals is permissible. Alternatively, the form-fill-seal accessory packet may be constructed as a three seamed pillow pouch, utilizing a back seam which joins the opposite edges of the packaging materials into a tube. The back seam shall be a fin seal which is formed by heat sealing the two opposite inside edges to each other and folding the fin seal flat in the plane of the material. The tube may be formed around the contents being packed, or it may be formed empty and filled in a subsequent step. The material specified in 3.1.2.1 shall be used. Tear nicks, notches, or serrations shall be provided on one or both end seals.

3.1.3 Ration components. The components required for ration assembly shall be as specified in table I.

TABLE I. Ration components

Component	Document reference	Used in ration No.
Beef stew	MIL-B-44140	2
Chicken a la king	MIL-C-44137	4
Chicken and rice	MIL-C-44138	5
Chicken stew	MIL-C-44148	1
Escalloped potatoes with pork	MIL-E-44141	3
Vermicelli with meat and sauce	MIL-V-44139	6
Soup, instant, reduced sodium, chicken noodle	MIL-S-44283	All rations
Oatmeal, instant	MIL-O-44278	
	Type I	2, 3
	Type II	4, 6
	Type III	1, 5
Brownies, chocolate covered	MIL-C-44072	2, 3, 5
Candy and chocolate confections,	A-A-20177	All rations
two 1/2 oz. toffee rolls, chocolate flavored	Type II	
one 1-1/2 oz. package pan-coated, chocolate disks	Type VI	
Cookies, oatmeal; chocolate covered	MIL-C-44072	1, 4, 6

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TABLE I. Ration components (cont'd)

Component	Document reference	Used in ration No.
Fruit bars	MIL-F-44279	
	Type I	2, 3, 4
	Type II	1, 5, 6
Oatmeal cookie bar and granola bar	MIL-O-44136	All rations
Nut raisin mix	MIL-N-44280	All rations
Beverage base, apple cider	MIL-B-44281	All rations
Beverage base: orange compressed	MIL-B-44135	All rations
Cocoa, beverage, powder	MIL-C-3031 Type I, class 1	All rations
Tea, instant, sweetened, lemon flavored	MIL-T-44284	All rations
Spoon, picnic, plastic	L-F-560 Type IV, item 9	All rations
Accessory packet	(see 3.1.4)	All rations

3.1.4 Accessory packet components. The components required for the assembly of the accessory packet shall be as specified in table II.

TABLE II. Accessory packet components

Component	Document reference	Quantity
Coffee, instant	HHH-C-575, type III, style A, blend (1) composition (1)	1 each
Cream substitute	MIL-C-43338	1 each
Sugar	A-A-20135, type I, class 1, 6 gm envelope	1 each
Chewing gum	A-A-20175, type I, flavors 1 or 2	2 tblts/pkg
Matches	EE-M-101, type II, class 2	1 book
Toilet tissue	MIL-P-43988	2 pkts
Closure device	Flip clip fastener 1/	2 each

1/ Use a "flip clip" fastener or a similarly constructed fastener. The "flip clip" fastener is manufactured by Avery Dennison Manufacturing Company, Framingham, MA 01701.

3.2 Accessory packet assembly. The required components as indicated in table II shall be inserted in an accessory packet as specified in 3.1.2. When a preformed packet is used, a minimum 1/8 inch wide heat seal shall be applied to effect closure of the packet. When a form-fill-seal packet is used, components shall be placed in the tray-shaped body and the cover applied by heat sealing (see 3.1.2.2). The average seal strength of the packet seals shall not be less than 3.5 pounds per inch of width and no individual specimen shall have a seal strength of less than 3.0 pounds per inch of width when tested in accordance with 4.2.2.2.

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observations, inspections, and investigations.

3.3 Ration bag assembly. The required components as specified in tables I and III shall be assembled and inserted into the appropriate ration bags A and B so as to occupy the minimum amount of space while maintaining a flat configuration. Excess air shall be expelled prior to closure sealing with heat such as to allow packing of the ration bags without damaging the ration bag or components therein. The closure seal shall have an average seal strength of not less than 9 pounds per inch of width and no individual specimen shall have a seal strength of less than 7 pounds (see 4.2.4.2). The two bags per ration (A+B) shall be taped together by wrapping a length of tape completely around the two bags girthwise. The tape shall be provided with a minimum 1/2 inch pull tab at the terminal end of the wrapping tape. The tape tab shall be made by folding the adhesive surface of the tape back to contact itself for a minimum length of 1/2 inch. The tape shall be 1/2 inch wide filament tape conforming to PPP-T-97, type II, class A. In lieu of folding tape over to provide a tab, a tab may be provided by the tape manufacturer by omitting the application of adhesive on a section of one end of each strip of tape on a pre-cut roll. As an alternative to taping, the menu bags may be banded together with a rubber band. The rubber band, when flaccid, shall be 3 inches in diameter, 3/8 inch wide (+ 1/16 inch) and shall be of a gauge thickness so that there are no more than 150 bands per pound (approx. 0.05 inch). The band shall be applied so as to bisect the length dimension of the menu bags into two equal halves.

TABLE III. Ration contents

Component	Ration No. 1	2	3	4	5	6	Ration bag	Qty in each ration 1/
Beef stew		X					B	1
Chicken a la king				X			B	1
Chicken and rice					X		B	1
Chicken stew	X						B	1
Escalloped potatoes with pork			X				B	1
Vermicelli with meat and sauce						X	B	1
Soup, instant and reduced sodium, chicken noodle	X	X	X	X	X	X	A	1
Oatmeal (apple & cinnamon)		X	X				A	1
Oatmeal (maple & brown sugar)				X		X	A	1
Oatmeal (strawberry)	X				X		A	1
Brownies, chocolate covered		X	X		X		B	1
Candy, toffee rolls, chocolate flavored	X	X	X	X	X	X	B	1
Candy, pan-coated, chocolate disks	X	X	X	X	X	X	B	1
Cookies, oatmeal, chocolate covered	X			X		X	B	1
Fruit bars, blueberry	X				X	X	A	1 2/
Fruit bars, fig		X	X	X			A	1 2/
Granola bars and oatmeal cookie bars	X	X	X	X	X	X	B	2 2/
Nut raisin mix	X	X	X	X	X	X	A	1
Beverage base, apple cider	X	X	X	X	X	X	A	1
Beverage base: orange, compressed	X	X	X	X	X	X	B	1
Cocoa beverage powder	X	X	X	X	X	X	A	2
Tea, instant, sweetened, lemon flavored	X	X	X	X	X	X	B	2

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TABLE III. Ration contents (cont'd)

Component	Ration No. 1	2	3	4	5	6	Ration bag	Qty in each ration 1/
Spoon, picnic, plastic	X	X	X	X	X	X	A & B	2
Accessory packet	X	X	X	X	X	X	A	1

1/ Quantity expressed in pouches, envelopes and packets.

2/ Two bars in each pouch.

3.4 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Packing or Holding Human Food", and the plant sanitation requirements of the appropriate Government inspection agency.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.1.1 Material certification. A certificate of compliance may be accepted as evidence that the characteristics listed below conform to the specified requirements.

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Material requirement	Requirement paragraph	Test procedure
Ration bag:		
Identification and thickness	3.1.1	Laboratory evaluation.
Color	3.1.1	Visual evaluation.
Accessory packet:		
Identification	3.1.2.1 and 3.1.2.2	Laboratory evaluation.
Aluminum foil thickness	3.1.2.1 and 3.1.2.2	As specified in QQ-A-1876.
Polyethylene, linear low density polyethylene, polypropylene, and polyester film thickness	3.1.2.1 and 3.1.2.2	As specified in L-P-378, except that a machinist's micrometer may be used provided that its graduations and accuracy conform to the requirements of L-P-378.
Color	3.1.2.1 and 3.1.2.2	Visual evaluation.

4.2.1.2 Unfilled ration bag and preformed accessory packet examination. A certificate of compliance may be accepted as evidence that unfilled bags and packets conform to the requirements specified in 3.1.1 and 3.1.2.1, respectively. When deemed necessary by the inspection agency, testing of the unfilled bags and packets for seal strength of manufacturer's seals shall be as specified in 4.2.1.3 and 4.2.1.4.

4.2.1.3 Preformed accessory packet seal strength testing. The unfilled preformed accessory packets shall be tested for seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One-half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. The lot size shall be expressed in packets. The sample unit shall be one unfilled preformed accessory packet. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Three adjacent specimens, 1/2 or 1 inch wide, shall be cut from each of the three sealed sides of each packet in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens. Any test

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specimen or average seal strength failing to meet the seal strength requirements specified in 3.1.2.1 shall be classified as a major defect.

4.2.1.4 Ration bag seal strength testing. The unfilled ration bags shall be tested for seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One-half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. The lot size shall be expressed in bags. The sample unit shall be one unfilled ration bag. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. Three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the sealed side of each bag in the sample. The results shall be reported to the nearest 0.1 pound per inch of width. The average seal strength of each seal shall be calculated by averaging the test results of the three test specimens. Any test specimen or average seal strength failing to meet the seal strength requirements specified in 3.1.1 shall be classified as a major defect.

4.2.2 Accessory packet inspection.

4.2.2.1 Accessory packet examination. The filled and sealed preformed or form-fill-seal packets shall be examined for the defects listed in table IV. The lot size shall be expressed in packets. The sample unit shall be one filled and sealed packet. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects.

TABLE IV. Accessory packet defects

Category		Defect
Major	Minor	
101		Not clean 1/
	201	Effective heat seal closure width less than 1/16 inch
	202	Tear nick, notch or serrations missing; or do not facilitate easy opening; or assembler's seal extends into or below tear nick or notch
	203	Dimensions not as specified
	204	Tear, hole, or open seal
	205	Evidence of delamination
	206	Labeling missing, incorrect, or illegible

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1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause package damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one package face or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one package face.

4.2.2.2 Accessory packet closure seal testing. The filled and sealed accessory packets shall be tested in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. Test specimens shall be cut to a length suitable for proper mounting. For preformed packets, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the closure seal of each packet in the sample. For form-fill-seal packets, three adjacent specimens, 1/2 or 1 inch wide, shall be cut from each side and each end of each packet in the sample. The average seal strength of the closure seal for the preformed packet shall be calculated by averaging the test results of the three test specimens cut from that seal. For the form-fill-seal packet, the average seal strength of each side seal and end seal of the packet shall be calculated by averaging the test results of the three specimens cut from each side seal or end seal. The results shall be reported to the nearest 0.1 pound per inch of width. The lot size shall be expressed in packets. The sample unit shall be one sealed packet. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5. Any test specimen or average seal strength failing to meet the requirements of 3.2 shall be classified as a major defect.

4.2.2.3 Accessory packet and contents examination. The accessory packet and contents of the filled and sealed packets shall be examined for the defects listed in table V (this examination may be performed on the preformed packets after filling and prior to sealing). The lot size shall be expressed in packets. The sample unit shall be one packet open or sealed. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 1.5 for major defects and 4.0 for minor defects.

TABLE V. Accessory packet and contents defects

Category		Defect
Major	Minor	
101		Missing component(s)

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TABLE V. Accessory packet and contents defects (cont'd)

Category		Defect
Major	Minor	
102		Foreign odor
103		Component not clean 1/
	201	Missing or unserviceable component (for example, tear, hole, or open seam in coffee, cream substitute, sugar, or crushed gum)

1/ Outer packaging shall be free from foreign matter which is unwholesome, has the potential to cause package damage (for example, glass, metal filings, etc.) or generally detracts from the clean appearance of the package. The following examples shall not be scored as defects for unclean:

a. Foreign matter which presents no health hazard or potential package damage and which can be readily removed by gently shaking the package or by gently brushing the package with a clean dry cloth.

b. Localized dried product which affects less than 1/8 of the total surface area of one package face or an aggregate of scattered dried product which affects less than 1/4 of the total surface area of one package face.

4.2.3 Shipping container inspection.

4.2.3.1 Shipping container examination. Examination of the filled and closed shipping containers shall be in accordance with the examination criteria of the appendix to PPP-B-636 except that the closure examination shall be in accordance with 4.2.3.3. In addition, the following defects shall be included in the table of examination:

Major: National stock number, item description, contract number or date of pack markings missing, incorrect, or illegible.
Metal fasteners used in closing the box.
Outer flaps do not completely meet, leaving an opening greater than 3/4 inch between flaps.

Minor: Other required markings missing, incorrect, or illegible.

4.2.3.2 Examination of adhered sleeve joint. A certificate of compliance may be accepted as evidence that the sleeve glue joint conforms to the requirements for weather-resistant boxes, glue closure examination, as specified in the appendix of PPP-B-636.

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4.2.3.3 Flap closure testing. The closure of the four outer flaps of the container shall be tested separately. A 90° angular bar with each leg approximately 5 inches long by 3 inches wide by 1/8 inch thick shall be used to test the flap closures. Insert one leg of the angular bar full length under the center of one outer flap. Insertion shall be made through the open slot between the outer flaps. Lift the container vertically by the other leg of the bar until the container is suspended. The complete upper surface of the inserted leg shall be in contact with the inner surface of the flap during the lifting and suspension of the container. Complete separation of the adhesive bond of one or more of the outer flaps, showing no evidence of fiber tear, shall be classified as a major defect. The lot size shall be expressed in shipping containers. The sample unit shall be one shipping container. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.3.4 Shipping container interior examination. The filled and sealed shipping containers shall be examined for defects listed in table VI. The lot size shall be expressed in shipping containers. The sample unit shall be one filled and sealed shipping container. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for the major defects and 10.0 for minor defects.

TABLE VI. Shipping container interior defects

Category		Defect
Major	Minor	
101		Less than six rations or not one of each
102		Ration bags not bound in ration units (see 3.3).
	201	Not packed or arranged as specified
	202	Box liner missing

4.2.4 Ration bag inspection.

4.2.4.1 Ration bag examination. The filled and sealed ration bags shall be examined for the defects listed in table VII. The lot size shall be expressed in bags. The sample unit shall be one filled and sealed ration bag. The sample shall consist of an equal number of bags A and B. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 4.0 for minor defects. The sample ration bags shall be selected from shipping containers which have been filled and closed. The inspection sample shall contain a proportionate amount of each of the six rations.

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TABLE VII. Ration bag defects 1/

Category		Defect
Major	Minor	
101		Ration component missing or incorrect assortment for ration
102		Not clean - the ration bag or any of the outer packaging of its contents 2/
103		Tear, hole, open seal, or puncture, in ration component package (applicable to instant oatmeal, fruit bars, candy, beverages, or soup packages)
104		Both spoons broken
105		Loss of vacuum (applicable to entree, oatmeal cookie and granola bar, chocolate flavored toffee roll, nut raisin mix, chocolate covered oatmeal cookie and brownie packages. Also applicable to chicken flavored noodle instant soup, when vacuum sealed.)
106		Labeling missing, incorrect or illegible
107		Foreign odor
108		Crushed (crumbled) or broken granola bars or oatmeal cookie bars, chocolate covered cookie or brownie, or other packaged components with damage adversely affecting its contents 3/
	201	One spoon broken
	202	Tear, hole, open seal, or split in ration bag
	203	Tear, hole, open seal, or puncture in accessory packet
	204	Tear notch or slit missing or not as specified

1/ Examination of ration bags shall be made after shipping container is filled, closed, sleeved, and reinforced (see 4.2.3).

2/ The following examples shall not be scored as defects for unclean:

a. Water spots.
b. Two or fewer specks of dried product each of which measure 1/8 square inch or less.

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c. Any foreign matter which presents no health hazard or potential packet damage and which readily falls off when packet is lifted and shaken lightly.
d. Very thin film of grease, oil or product residue which is discernible to touch, but not readily discernible by visual examination.

3/ Examination for crushed products shall be performed on ciosed component packages. A package containing crushed or broken components shall be indicated by a depression or bump when the package is passed through the fingers while exerting sufficient finger grip pressure to feel the serrated areas, by a movement of any particles felt through the package, or by a permanent wrinkling of the package material over the ration component. Also, crushed component is one in which 1/8 or more of the volume of the product has been reduced to crumbs or powder.

4.2.4.2 Ration bag closure seal strength testing. The filled and sealed ration bags shall be tested for closure seal strength in accordance with ASTM F 88, except that the specimen holding clamps shall be spaced 2 inches apart prior to testing and the testing speed shall be 10 or 12 inches per minute. Machines that apply the tensile load to the test specimen by movement of the upper or lower clamp may be used. One-half or 1 inch wide test specimens shall be cut to a length suitable for proper mounting. Three adjacent specimens, 1/2 or 1 inch wide, shall be cut from the closure seal of each bag in the sample. The results shall be reported to the nearest 0.1 pounds per inch of width for each specimen and for the average of the three specimens from each closure seal. Any individual specimen or any closure seal average failing to meet the seal strength requirement specified in 3.3 shall be classified as a major defect. The lot size shall be expressed in bags. The sample unit shall be one bag of one filled and sealed ration. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

4.2.5 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Packing. Six rations, one of each from table III shall be packed in a fiberboard box with box liner and sleeve constructed and closed in accordance with style RSC-L-SL, grade V2s of PPP-B-636, except metal fasteners shall not be used in the closure of the flaps, the gap between either flaps shall not be more than 1/2 inch wide, and adequacy of the closure shall be determined by testing as specified in 4.2.3.3. The rations, assembled as specified in 3.3, shall be packed on edge, two rations in length, three rations in width, and one ration in depth. Alternative packing arrangements may be used provided that the rations are packed along the lengthwise edge and that the two ration bags of each ration remain secured together. The inside of each shipping container shall be fitted with a box liner conforming to grade V3c of PPP-B-636. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The liner shall be constructed with a butt joint and secured with tape conforming

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to PPP-T-76. An overlapped glued joint located at the center of a side panel or in a corner of the box liner may be used in lieu of the taped joint. The amount of overlap shall not be less than 1-1/4 inches and shall extend along the full height of the box liner. Inner flaps of the box may be closed with two strips of hot melt adhesive per flap, provided the closure meets the test requirements of 4.2.3.3. As an alternative to body joint metal fasteners, adhesive conforming to MMM-A-250, applied in accordance with "adhesive joint" requirement of PPP-B-636, may be used to secure the manufacturer's joint of the V2s box. As an alternative to sleeve joint metal fasteners, adhesive conforming to MMM-A-250 may be used for securing the joint, except the overlap shall be a minimum of 3 inches wide. The adhesive joint shall be examined in accordance with 4.2.3.2. The adhesive shall cover a minimum of 90 percent of the sleeve overlap area. Sleeved boxes shall be reinforced in accordance with the appendix of PPP-B-636, except pressure-sensitive adhesive filament-reinforced tape is prohibited and boxes shall be reinforced with two girthwise straps positioned to divide the box into units of equal length. The inside dimensions of the fiberboard box shall be 19 inches in length, 14-3/4 inches in width, and 5-5/8 inches in depth.

5.2 Shipment of accessory packets to ration assembler. Accessory packets shall be packed in containers which are acceptable to common carriers for safe transportation to the destination specified in shipping instructions at the lowest rate for such supplies. Shipping containers shall be in accordance with the Uniform Freight Classification or National Motor Freight Classification, as applicable. Marking shall shall be in accordance with MIL-STD-129 and as specified in the contract.

5.3 Unit loads. The ration, cold weather, packed as specified in 5.1, shall be arranged in unit loads in accordance with type I, class B of MIL-L-35078.

5.4 Labeling and marking.

5.4.1 Ration bags. Each ration bag shall be clearly printed on at least one face in permanent black letters not less than 3/16 inch high as follows:

RATION, COLD WEATHER
RATION NO. (1A through 6B as applicable)
List of components in each bag
Name and address of assembler

5.4.2 Shipping containers. Shipping containers for complete ration, cold weather shall be marked in accordance with MIL-STD-129 and as specified in the contract with the following exceptions:

a. Identification markings normally placed on an end of the shipping container shall be placed on the corresponding end of the sleeve. Contract data and other required markings shall be placed on the side of the shipping container to the right of the marked end of the sleeve (when sleeve is positioned on the shipping container).

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b. The top of the sleeve shall be marked "Ration, Cold Weather" in letters not less than 1-1/4 inches high.

c. The bottom end of the sleeve as well as the bottom of the case shall convey the box maker's compliance markings as required.

5.4.3 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078 and as specified in the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When accessory packets are procured by the government, the acquisition agency should specify shipment to the assembly point as indicated in 5.2.

6.2 Subject term (key word) listing.

Combat field feeding
Operational rations

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - GL

Review activities:

Army - MD, QM
Navy - MC, SA
Air Force - 50
DP - SS

Preparing activity:

Army - GL

(Project 8970-0164)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-44277A(GL)	1990 November 15

3. DOCUMENT TITLE

RATION, COLD WEATHER, ASSEMBLY OF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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